E-COMMERCE SALES DATASET

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1. Dataset Overview

This documentation describes the structure, schema, and content of the ecommerce_sales_dataset.xlsx workbook. The dataset captures transactional and operational data for an Indian e-commerce business operating across multiple platforms and cities.

1.1 File Summary

Metric	Value
File Name	ecommerce_sales_dataset.xlsx
Format	Microsoft Excel Workbook (.xlsx)
Total Sheets	7
Total Records (master_data)	200 rows
Geographic Coverage	India (Delhi, Chennai, Hyderabad, Bangalore, Mumbai)
Platforms Covered	AJIO, Snapdeal, Flipkart, Amazon
Product Categories	Fashion, Beauty, Electronics, Home
Time Period	Full calendar year (January — December)
Currency	INR (Indian Rupees)

1.2 Workbook Structure

Metric	Value
Sheet Name	Purpose
master_data	Primary transactional data — 200 orders with full financial and monthly sales breakdown
Input_data1	Vendor-level aggregated data with return type and city quantity metrics
Input_data3	Monthly sales data enriched with geographic and vendor information
input_data	Derived calculations — annual totals, profit, and profit percentage formulas
pivot_tables	Pre-built pivot summaries for amount, quantity, profit and monthly sales by city
Analysis	Empty sheet reserved for analysis output
dash_bord	Empty sheet reserved for dashboard visualizations

2. Sheet: master_data

The master_data sheet is the primary data source for the workbook. It contains 200 order records capturing customer, product, financial, and monthly sales information for each transaction.

2.1 Sheet Dimensions

Metric	Value
Rows	200 (excluding header)
Columns	26
Key Identifier	Order_ID (e.g., ORD1000 — ORD1199)

2.2 Field Definitions

2.2.1 Order & Customer Fields

Field Name	Data Type	Description
`Order_ID`	String	Unique identifier for each order. Format: ORD followed by 4-digit number (e.g., ORD1000).
`Customer_ID`	String	Unique customer identifier. Format: CUST followed by 3-digit number (e.g., CUST202).
`City`	String	City where the order was placed. Values: Delhi, Chennai, Hyderabad, Bangalore, Mumbai.
`Category`	String	Product category. Values: Fashion, Beauty, Electronics, Home.
`Product_Name`	String	Name of the product ordered (e.g., Product_A, Product_B, Product_C, Product_D).
`ORD1000`	Integer	Quantity of units ordered per transaction. Observed range: 1 to 100.
`Platform`	String	E-commerce platform used. Values: AJIO, Snapdeal, Flipkart, Amazon.
`Payment_Method`	String	Payment method used. Values: Credit Card, Debit Card, UPI, Cash on Delivery.

2.2.2 Financial Fields

Field Name	Data Type	Description
`Unit_Price`	Decimal (INR)	Price per unit before discount. Range: INR 15.98 — INR 498.87. Mean: INR 263.11.

Field Name	Data Type	Description
`Discount (%)`	Decimal (%)	Percentage discount applied to the order. Range: 0.01% — 29.98%. Mean: 15.48%.
`Revenue`	Decimal (INR)	Total revenue generated from the order after discount. Range: INR 105 — INR 4,952. Mean: INR 2,563.
`Cost`	Decimal (INR)	Total cost of goods for the order. Range: INR 76.64 — INR 2,980.34. Mean: INR 1,543.
`Profit`	Decimal (INR)	Net profit per order (Revenue minus Cost). Range: INR 50.65 — INR 1,991.19. Mean: INR 973.
`Delivery_Time (Days)`	Integer	Number of days taken for delivery. Range: 1 — 9 days. Mean: 5.1 days.

2.2.3 Monthly Sales Fields (Jan — Dec)

Columns Jan_Sales through Dec_Sales capture the monthly sales figures for each order record across the full calendar year. All values are integers representing INR sales amounts.

Field Name	Data Type	Description
`Jan_Sales`	Integer (INR)	Sales amount recorded for January.
`Feb_Sales`	Integer (INR)	Sales amount recorded for February.
`Mar_Sales`	Integer (INR)	Sales amount recorded for March.
`Apr_Sales`	Integer (INR)	Sales amount recorded for April.
`May_Sales`	Integer (INR)	Sales amount recorded for May.
`Jun_Sales`	Integer (INR)	Sales amount recorded for June.
`Jul_Sales`	Integer (INR)	Sales amount recorded for July.
`Aug_Sales`	Integer (INR)	Sales amount recorded for August.
`Sep_Sales`	Integer (INR)	Sales amount recorded for September.
`Oct_Sales`	Integer (INR)	Sales amount recorded for October.
`Nov_Sales`	Integer (INR)	Sales amount recorded for November.
`Dec_Sales`	Integer (INR)	Sales amount recorded for December.

2.3 Dataset Summary Statistics

Metric	Value
Total Orders	200
Total Revenue	INR 5,12,641.11

Metric	Value
Total Cost	INR 3,08,619.49
Total Profit	INR 1,94,667.48
Average Revenue per Order	INR 2,563.21
Average Profit per Order	INR 973.34
Average Discount Applied	15.48%
Average Delivery Time	5.1 days

2.4 Category Distribution

Metric	Value
Electronics	65 orders (32.5%)
Beauty	46 orders (23.0%)
Home	45 orders (22.5%)
Fashion	44 orders (22.0%)

2.5 Platform Distribution

Metric	Value
Snapdeal	59 orders (29.5%)
AJIO	49 orders (24.5%)
Flipkart	47 orders (23.5%)
Amazon	45 orders (22.5%)

2.6 City Distribution

Metric	Value
Chennai	44 orders (22.0%)
Bangalore	43 orders (21.5%)
Mumbai	41 orders (20.5%)
Hyderabad	39 orders (19.5%)
Delhi	33 orders (16.5%)

3. Sheet: Input_data1

The Input_data1 sheet contains vendor-level aggregated data. Each row corresponds to a vendor (customer) and includes transaction metadata, quantity sold, return type, and total profit. This sheet is used as an input source for pivot analysis.

3.1 Sheet Dimensions

Metric	Value
Rows	200 (excluding header)
Columns	10
Key Identifier	vendors_ID (maps to Customer_ID in master_data)

3.2 Field Definitions

Field Name	Data Type	Description
`vendors_ID`	String	Vendor identifier. Corresponds to Customer_ID in master_data (e.g., CUST202).
`Platform`	String	E-commerce platform. Values: AJIO, Snapdeal, Flipkart, Amazon.
`Item_Type`	String	Product category. Values: Fashion, Beauty, Electronics, Home.
`Cost`	Decimal (INR)	Cost of goods for the vendor transaction.
`Quantity`	Integer	Number of units associated with the vendor record.
`Payment_Method`	String	Payment method used. Values: Credit Card, Debit Card, UPI, Cash on Delivery.
`Return_Type`	String	Type of return/resolution. Values: Exchanges, Refunds, Replacements, Warranties.
`City`	String	City of the vendor. Values: Delhi, Chennai, Hyderabad, Bangalore, Mumbai.
`Quantity_city`	Integer	Total quantity sold in the vendor's city. Range observed: ~10,500 — ~10,700.
`Profit`	Decimal (INR)	Total profit attributed to the vendor. Range: INR ~50K — INR ~3.5M.

3.3 Return Type Distribution

Metric	Value
Exchanges	63 records (31.5%)
Warranties	47 records (23.5%)
Replacements	46 records (23.0%)
Refunds	44 records (22.0%)

4. Sheet: Input_data3

The Input_data3 sheet extends the vendor data from Input_data1 by splitting monthly sales into individual monthly columns and adding full geographic context including country and state fields.

4.1 Sheet Dimensions

Metric	Value
Rows	200 (excluding header)
Columns	24
Key Identifier	vendors_ID

4.2 Field Definitions

Field Name	Data Type	Description
`jan – dec`	Integer (INR)	Monthly sales values for each calendar month (January through December).
`country`	String	Country of the vendor. All values are 'india' in this dataset.
`City`	String	City of the vendor. Values: Delhi, Chennai, Hyderabad, Bangalore, Mumbai.
`state`	String	State corresponding to the city (e.g., Delhi, Tamil Nadu, Telangana, Karnataka, Maharashtra).
`Platform`	String	E-commerce platform. Values: AJIO, Snapdeal, Flipkart, Amazon.
`Payment_Method`	String	Payment method used by the vendor.
`vendors_ID`	String	Vendor identifier matching master_data and Input_data1.
`Item_Type`	String	Product category for this vendor record.
`Cost`	Decimal (INR)	Cost of goods for this vendor transaction.
`unit_Quantity`	Integer	Unit quantity for this vendor record.
`Return_Type`	String	Type of return/resolution. Same values as Input_data1.
`Quantity_city`	Integer	Total quantity sold in the vendor's city.
`Profit`	Decimal (INR)	Total profit for the vendor. Matches Input_data1 Profit column.

4.3 Geographic Mapping

Metric	Value
Delhi	Delhi (State / UT)
Chennai	Tamil Nadu
Hyderabad	Telangana
Bangalore	Karnataka
Mumbai	Maharashtra

5. Sheet: pivot_tables

The pivot_tables sheet contains pre-computed summary tables created from the source data. It consolidates key metrics by city and return type for quick reference and dashboard use.

5.1 Pivot Summary: Monthly Sales by City

Aggregates monthly sales (Jan — Dec) broken down by city. All values are in INR.

Metric	Value
City	Annual Sales (Sum Jan–Dec, INR)
Bangalore	~14,556,974
Chennai	~13,181,219
Hyderabad	~12,343,039
Mumbai	~13,507,959
Delhi	~9,936,112

5.2 Other Pivot Metrics

Metric	Value
Sum of Amount (Total)	62,426,129
Sum of Quantity (Total)	10,356
Sum of Profit (Total)	15,606,532.25

5.3 Quantity by City

A secondary pivot within this sheet summarises total quantity sold per city (Bangalore, Chennai, Delhi, Hyderabad, Mumbai) and by return type (Exchanges, Refunds, Replacements, Warranties).

6. Sheet: input_data

The input_data sheet contains derived calculation columns. It includes annual sales totals (sum of Jan — Dec), total profit, and profit percentage calculations using the formula: Profit % = Profit / (Sales - Profit).

This sheet is structured with wide multi-column formulas spanning 48 columns and is intended for formula-based computation rather than raw data entry.

7. Sheets: Analysis & dash_bord

Both the Analysis and dash_bord sheets are currently empty. They are reserved as output destinations for analytical summaries and interactive dashboard visualizations built on top of the source data sheets.

8. Data Relationships & Key Joins

The following fields act as join keys across sheets in the workbook:

Source Sheet	Target Sheet	Join Key
master_data	Input_data1	`Customer_ID = vendors_ID`
master_data	Input_data3	`Customer_ID = vendors_ID`
Input_data1	Input_data3	`vendors_ID (same field, both sheets)`
master_data	pivot_tables	`City (aggregated in pivot)`
Input_data1	pivot_tables	`Return_Type, City, Quantity_city`

8.1 Notes on Data Integrity

- Order_ID values in master_data appear to be unique per row and follow a sequential format (ORD1000 — ORD1199).
- vendors_ID in Input_data1 and Input_data3 maps directly to Customer_ID in master_data.
- The ORD1000 column in master_data appears to store quantity values (1–100) despite being named after an order ID — treat this as a quantity field.
- Profit values in Input_data1 and Input_data3 represent aggregated vendor-level profit totals, which are significantly larger than per-order Profit values in master_data.
- All monetary values are in Indian Rupees (INR). No currency conversion is applied.

9. Suggested Use Cases

9.1 Sales Performance Analysis

- Compare monthly sales trends across the 12-month period using the Jan_Sales — Dec_Sales columns.
- Identify peak and off-peak months by city or platform.
- Measure revenue vs. profit margins by category and platform.

9.2 Platform & Vendor Benchmarking

- Compare order volume and average revenue across AJIO, Snapdeal, Flipkart, and Amazon.
- Assess vendor-level profitability using Input_data1 and Input_data3 Profit fields.
- Evaluate average discount offered per platform.

9.3 Customer & Geographic Insights

- Analyse spending patterns and delivery times by city.
- Use Input_data3 state mapping to perform state-level geographic analysis.
- Identify high-value cities using the pivot_tables monthly sales aggregates.

9.4 Returns & After-Sales Analysis

- Use the Return_Type field in Input_data1 and Input_data3 to understand post-purchase behaviour.
- Break down returns by city, platform, and item type.
- Calculate return rate as a percentage of total orders.

9.5 Dashboard & Reporting

- Use the pivot_tables sheet as a data source for charts in the dash_bord sheet.
- Build category-level revenue and profit dashboards using master_data.
- Use the input_data sheet formulas as the basis for automated profit percentage KPI cards.